Submission Summary

Conference Name

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3629

Paper Title

An Implemented Blockchain-Based Framework for Healthcare Data Sharing

Abstract

Health care is one of the largest generators of personal, private, and sensitive data in the world. With this large volume of sensitive patient information being exchanged securely across numerous entities such as hospitals, labs, insurance companies, and most importantly the patient themselves. Centralized systems used to protect patient information have many shortcomings including but not limited to: Data breaches; Limited interoperability; Lack of transparency in who can see what; Patient's lack of control regarding their Personal Health Record (PHR). As a result, there is an increased interest in using a new type of distributed ledger system called "Blockchain" which is immutable, public, and allows for peer-to-peer transactions without needing a central authority. The goal of this research is to provide a review of all existing healthcare-related data sharing models utilizing the blockchain architecture. Identify their respective shortcomings. Provide a conceptual model that focuses on the needs of the patient. Use smart contracts for access control and use off-chain encrypted storage for scalable data management. The proposed model increases the data integrity. Provides auditable trail. Empowers the patient to directly manage their medical record and decreases dependency on centralized authorities. An experimental analysis will compare the performance of the proposed model against other similar models to show that it is viable for actual deployment within real-world healthcare environments

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Submission Files

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